

Module 7: Date Functions & Number Formatting

DEVELOPED BY TALENCIAGLOBAL

MODULE 7

Duration: 26 minutes | **Learn:** 8 min | **Lab:** 18 min

Dates are numbers in disguise. Once you understand that, every date calculation becomes simple arithmetic.

Learn

Core date functions, serial numbers, and custom formatting concepts

Practice

Hands-on lab with real HR and financial datasets

Apply

Industry-ready formulas for tenure, deadlines, and Indian financial reporting

How Excel Stores Dates

Behind every date in Excel is a serial number. Excel stores dates as sequential integers beginning from January 1, 1900 (serial number 1). Every subsequent day increments the count by exactly one — making date arithmetic as straightforward as addition and subtraction.

Date	Serial Number
1-Jan-1900	1
1-Jan-2000	36,526
1-Jan-2025	45,658
10-Jun-2026	46,178

Why This Matters

- Subtracting two dates yields the number of days between them: `=B2-A2`
- Adding a number to a date produces a future date: `=TODAY()+30`
- Functions like DATEDIF, EDATE, and NETWORKDAYS all rely on this numeric foundation

Common Trap

If a date displays as a large number (e.g., 45658) instead of a recognisable date, the cell is formatted as **General** or **Number** rather than **Date**.



Right-click → Format Cells → Date to restore the correct display. The underlying value is always intact.

Current Date & Time: TODAY and NOW

Two functions that always return the current moment. Neither function accepts any arguments — the parentheses are always empty.

=TODAY()

Returns today's date with **no time component**.
Recalculates every time the workbook opens or recalculates.

=NOW()

Returns the current **date and time**. Updates on every recalculation — press F9 to observe the time component change.

Common Professional Uses



Age or Tenure Calculation

=DATEDIF(H2, TODAY(), "Y") →
years elapsed since join date



Overdue Flag

=IF(DueDate<TODAY(),
"Overdue", "On Track") → instant
SLA visibility



Report Timestamp

="Report generated on: " &
TEXT(NOW(), "DD-MMM-YYYY
HH:MM")



Volatile Functions Alert: TODAY() and NOW() recalculate on every workbook change. For a permanent, auditable timestamp, use **Ctrl+;** (static date) or **Ctrl+Shift+;** (static time) — these insert fixed values that never change.

Extracting Date Components

Pull apart any date into its constituent elements — year, month, day, and weekday — for grouping, filtering, and conditional logic in large datasets.

Function	Extracts	Example (10-Jun-2026)	Result
YEAR(date)	The year	=YEAR("10-Jun-2026")	2026
MONTH(date)	Month number (1–12)	=MONTH("10-Jun-2026")	6
DAY(date)	Day of month (1–31)	=DAY("10-Jun-2026")	10
WEEKDAY(date)	Day of week (1=Sun, 7=Sat)	=WEEKDAY("10-Jun-2026")	4 (Wednesday)

Practical Applications

- Group by year:** Add `=YEAR(H2)` as a helper column, then pivot by year for annual trend analysis
- Filter by month:** `=MONTH(H2)` → filter for 6 = June headcount reports
- Identify weekends:** `=IF(OR(WEEKDAY(H2)=1, WEEKDAY(H2)=7), "Weekend", "Weekday")`

Rebuilding a Date

Use the `DATE(year, month, day)` function to reconstruct a valid date from separate components:

`=DATE(2026, 6, 10)` → **10-Jun-2026**

This is essential when importing data where year, month, and day arrive in separate columns — a common scenario in ERP and HRMS exports.

Date Arithmetic: DATEDIF, EDATE & EOMONTH

The three power functions for professional date calculations. These are the workhorses of HR analytics, payroll processing, and financial planning — used daily across thousands of Indian enterprises.

1

DATEDIF

Calculates the difference between two dates in a specified unit.

`=DATEDIF(start_date, end_date, unit)`

- "Y" → Complete years of tenure
- "M" → Total complete months
- "D" → Total calendar days
- "YM" → Remaining months after full years
- "MD" → Remaining days after full months

2

EDATE

Moves a date forward or backward by a specified number of months — preserving the day of month.

`=EDATE(start_date, months)`

`=EDATE(H2, 12)` → The date exactly one year after joining — ideal for probation completion and annual review scheduling.

3

EOMONTH

Returns the last day of a month, offset by a given number of months.

`=EOMONTH(date, months_offset)`

- `=EOMONTH(TODAY(), 0)` → Last day of current month
- `=EOMONTH(TODAY(), 3)` → Last day of month 3 months ahead

Full Tenure String Formula: `=DATEDIF(H2,TODAY(),"Y") & " years, " & DATEDIF(H2,TODAY(),"YM") & " months"` → Produces output such as **"8 years, 3 months"** — the format business stakeholders expect in appraisal and compliance reports.

Working Days: NETWORKDAYS & WORKDAY

Count and calculate business days between dates, automatically excluding weekends and optionally excluding public holidays. In project management and HR operations, calendar-day calculations routinely overcount by approximately 40% — a critical error in SLA and compliance contexts.

NETWORKDAYS

`=NETWORKDAYS(start_date, end_date, [holidays])`

- `=NETWORKDAYS(H2, TODAY())` → Working days since join date
- `=NETWORKDAYS("1-Jan-2026", "31-Mar-2026")` → Working days in Q1 2026
- `=NETWORKDAYS("1-Jan-2026", "31-Mar-2026", AA1:AA15)` → Excludes weekends AND listed public holidays

WORKDAY

`=WORKDAY(start_date, days, [holidays])`

- `=WORKDAY(TODAY(), 10)` → Date 10 working days from today
- `=WORKDAY(TODAY(), 10, AA1:AA15)` → Also skips listed holidays

Key Insight

10 years = ~3,652 calendar days but only ~2,608 working days. Always use NETWORKDAYS for payroll, SLA, and project timelines.



Project Deadlines

"This task takes 15 working days" →
`=WORKDAY(StartDate, 15)`



SLA Tracking

"Response due within 5 business days"
→ compare `=WORKDAY(TicketDate, 5)`
with today



Payroll Calculation

"How many working days did the employee work this month?" →
NETWORKDAYS with month start and end

Custom Number Formatting

Change how numbers appear without altering their underlying value. Custom formats are applied via Right-click → Format Cells → Number → Custom. The formula bar always retains the true numeric value — only the display changes.

Format Code	Input Value	Displayed As	Use Case
<code>#,##0</code>	1500000	1,500,000	International standard
<code>#,##,##0</code>	1500000	15,00,000	Indian lakh format
<code>₹#,##,##0</code>	1500000	₹15,00,000	Indian financial reporting
<code>0.0%</code>	0.18	18.0%	Percentage display
<code>0.00</code>	3.5	3.50	Consistent decimal places
<code>#,##0;(#,##0);"-"</code>	-500 / 500 / 0	(500) / 500 / -	Accounting style
<code>0000</code>	45	0045	Zero-padded IDs

The Indian Number Format

The format code `#,##,##0` groups digits as: ones → hundreds → then groups of two (lakhs, crores). This is **essential for Indian financial reporting** where audiences think in lakhs and crores, not millions and billions.

Three-Section Format

Format codes support three sections separated by semicolons: **positive ; negative ; zero**

`₹#,##,##0;(₹#,##,##0);"-"`

- Positive: ₹15,00,000
- Negative: (₹15,00,000)
- Zero: -



Key Principle: Custom formatting only changes the display. The underlying value remains unchanged. `=A1*2` still calculates correctly even when A1 displays as "₹15,00,000".

Custom Date Formatting

Control exactly how dates appear across reports, dashboards, and concatenated strings. Consistent date formatting is a hallmark of professional-grade workbooks and eliminates regional ambiguity.

Format Code	Input	Displayed As
DD-MMM-YYYY	10-Jun-2026	10-Jun-2026
DD/MM/YYYY	10-Jun-2026	10/06/2026
MMMM YYYY	10-Jun-2026	June 2026
DDD, DD MMM	10-Jun-2026	Wed, 10 Jun
DDDD, DD MMMM YYYY	10-Jun-2026	Wednesday, 10 June 2026
YYYY-MM-DD	10-Jun-2026	2026-06-10 (ISO format)

Format Code Reference

Code	Meaning	Example
D / DD	Day	5 / 05
DDD / DDDD	Weekday	Wed / Wednesday
M / MM	Month number	6 / 06
MMM / MMMM	Month name	Jun / June
YY / YYYY	Year	26 / 2026
H / HH	Hour	3 / 03
MM (after H)	Minute	05
SS	Seconds	30

TEXT Function for Formulas

Use `TEXT()` when embedding a formatted date inside a concatenated string. Without it, the serial number appears instead of the date.

`=TEXT(H2, "DD-MMM-YYYY")` → "15-Mar-2020"

`= "Joined on " & TEXT(H2, "DDDD, DD MMMM YYYY")` → "Joined on Friday, 15 March 2020"

 Without `TEXT()`: "Report for 46178" — always use `TEXT()` when concatenating dates.

Best Practices

Working with Dates and Number Formats Professionally

1. Store Dates as Dates, Not Text

If `LEFT`, `RIGHT`, and `LEN` work on a date cell, it is stored as text.

Convert using `DATEVALUE()` or the Text to Columns wizard. Text-dates break every date function silently.

2. Use `DATEDIF` for Human-Readable Tenure

Simple subtraction yields days.

`DATEDIF` with `"Y"` and `"YM"` produces **"8 years, 3 months"** — the format expected in appraisals, compliance reports, and board presentations.

3. Use `NETWORKDAYS` for Business Timelines

Calendar days overcount by approximately 40% due to weekends. Any SLA, project deadline, or payroll calculation must use working days to remain accurate and defensible.

4. Apply Indian Number Format for Indian Financial Data

The international format (`#,##0`) groups by thousands — confusing for audiences who think in lakhs and crores. Always apply `₹#,##,##0` for Indian financial reporting.

5. Format Dates Consistently Across the Workbook

Select one format — **DD-MMM-YYYY** is the safest, being unambiguous across all regions — and apply it uniformly. Mixed formats are a leading cause of data interpretation errors in shared workbooks.

6. Never Hardcode Dates in Formulas

Use `TODAY()` for current-date references and cell references for specific dates. Hardcoded dates make workbooks static, difficult to audit, and prone to silent errors after the date passes.

7. Use `TEXT()` When Embedding Dates in Strings

Without `TEXT()`, concatenation exposes the serial number:
"Report for 46178" instead of *"Report for 10-Jun-2026"*. Always wrap date references in `TEXT()` inside string formulas.

Module 7 Summary

You now have complete command of Excel's date and number formatting capabilities — from the foundational serial number model to professional-grade Indian financial formatting. The following reference consolidates every key function covered in this module.

Date Foundations - Dates are serial numbers from 1-Jan-1900 - TODAY() and NOW() — volatile, dynamic - Ctrl+; and Ctrl+Shift+; — static timestamps	Component Functions - YEAR(), MONTH(), DAY() - WEEKDAY() for weekend detection - DATE(y,m,d) to rebuild dates
Date Arithmetic - DATEDIF() — tenure in Y, M, D - EDATE() — shift by months - EOMONTH() — end of month	Working Days - NETWORKDAYS() — count business days - WORKDAY() — find future business date - Holiday range as optional third argument

7

Core Functions

Date functions mastered in this module

40%

Overcount Risk

Calendar days overcount vs. working days over a 10-year period

26

Minutes

Total module duration including lab exercises

₹0

Value Change

Custom formatting never alters the underlying cell value

✔ **Next Module:** Data Validation, Dropdowns, and Conditional Formatting — building interactive, error-resistant workbooks for enterprise use. *Developed by Talenciaglobal.*